

Unit 5: Continuity

1. The idea, in plain words

A function is continuous at a point if you could trace its graph through that point without lifting your pencil off the paper. No holes, no sudden jumps, no shooting off to infinity — just a smooth, unbroken path.

In the last unit, we learned to check whether the left-hand and right-hand limits agree. That was step one. Continuity asks one more question on top of that: does the function's actual value at that point match what it's approaching?

Here's the full checklist — think of it like a 3-question inspection you run at a single point $x = a$:

1. Is $f(a)$ actually defined? (Is there even a dot on the graph there?)
2. Does the limit exist as $x \rightarrow a$? (Do the left and right limits agree — no jump?)
3. Does that limit actually equal $f(a)$? (Does the dot sit exactly where the two sides are heading — no hole, no mismatch?)

If all three answers are “yes,” the function is continuous at a . If even one fails, it's discontinuous there.

The three ways continuity can break, pictured:

- Removable discontinuity (a hole): the limit exists just fine, but either $f(a)$ is undefined, or it's defined to some other value that doesn't match the limit. Picture an open circle on the curve with a lonely dot sitting somewhere else (or no dot at all). You could “fix” this by simply redefining that one point.
- Jump discontinuity: the left and right limits exist but disagree with each other — like a staircase step. No single point-fix can repair this; the two sides are just heading toward different places.
- Infinite discontinuity: the function shoots off toward $+\infty$ or $-\infty$ near that point (usually because of a denominator hitting zero without anything canceling it). Picture a vertical asymptote.

2. Toolbox

The continuity checklist at $x = a$: f is continuous at a if and only if all three hold:

$$1. f(a) \text{ is defined} \quad 2. \lim_{x \rightarrow a} f(x) \text{ exists} \quad 3. \lim_{x \rightarrow a} f(x) = f(a)$$

Where continuity usually “just works” automatically: polynomials are continuous everywhere; rational functions (fractions of polynomials) are continuous everywhere except where the denominator is zero; square roots are continuous everywhere they're defined (inside the domain).

Solving for an unknown to force continuity at a breakpoint: set the left-hand piece's value/limit equal to the right-hand piece's value/limit at that breakpoint, then solve the resulting equation for the unknown.

Classifying a discontinuity:

- Limit exists, but doesn't match $f(a)$ (or $f(a)$ is undefined) $\rightarrow$ removable

- Left limit $\neq$ right limit $\rightarrow$ jump
- Function blows up toward $\pm\infty \rightarrow$ infinite

3. Common mistakes

- Checking only that the left and right limits agree, and stopping there. That only confirms the limit exists — you still have to check it matches $f(a)$.
- Forgetting to check that $f(a)$ is even defined in the first place. A function can't be continuous at a point where there's no dot at all.
- Assuming a piecewise function is automatically continuous everywhere just because each individual piece is a nice, continuous formula. You must still separately check every breakpoint where the pieces are glued together.
- Mixing up removable and jump discontinuities. A hole (removable) can be patched by re-defining one point; a jump cannot, because the two sides are approaching genuinely different values.
- Sign or algebra slips while solving for the unknown parameter. These problems usually reduce to a simple one-variable equation — take it slow.

4. Problem Set

Warm-up

1. Let $f(x) = x^2 + 3$. Check whether f is continuous at $x = 2$ by verifying all three conditions.
2. Let $f(x) = \frac{x+1}{x-2}$. Check continuity at $x = 0$ and at $x = 2$. Explain what happens at $x = 2$.
3. Let $f(x) = \begin{cases} x+2, & x < 1 \\ 4, & x \geq 1 \end{cases}$. Check continuity at $x = 1$.
4. Let $f(x) = \begin{cases} x^2, & x \leq 2 \\ 2x, & x > 2 \end{cases}$. Check continuity at $x = 2$.
5. Let $f(x) = \begin{cases} 7, & x = 0 \\ x^2 + 3, & x \neq 0 \end{cases}$. Check continuity at $x = 0$, and describe what kind of discontinuity (if any) occurs.
6. Let $f(x) = \frac{1}{x-3}$. Check continuity at $x = 3$, and describe what kind of discontinuity (if any) occurs.

Standard

7. Find the value of a that makes f continuous at $x = 3$, where $f(x) = \begin{cases} x^2 - 2x + 3, & x < 3 \\ 2ax, & x \geq 3 \end{cases}$.
8. Find the value of k that makes f continuous at $x = 2$, where $f(x) = \begin{cases} kx + 1, & x < 2 \\ x^2 - 1, & x \geq 2 \end{cases}$.
9. Find the value of b that makes f continuous at $x = 1$, where $f(x) = \begin{cases} 3x + b, & x \leq 1 \\ x^2 + 2, & x > 1 \end{cases}$.
10. Find the value of m that makes f continuous at $x = -1$, where $f(x) = \begin{cases} x^2 + 4, & x < -1 \\ mx - 3, & x \geq -1 \end{cases}$.

11. Find the value of a that makes f continuous at $x = 2$, where $f(x) = \begin{cases} ax^2, & x \leq 2 \\ 4x - 4, & x > 2 \end{cases}$.
12. Find the value of c that makes f continuous at $x = 1$, where $f(x) = \begin{cases} x^3, & x < 1 \\ c - x, & x \geq 1 \end{cases}$.

Challenge

13. Let $f(x) = \begin{cases} ax + b, & x < 0 \\ x^2 + 1, & 0 \leq x \leq 2 \\ 3x - 1, & x > 2 \end{cases}$. You're also told that $f(-2) = 7$. Find a and b so that f is continuous at $x = 0$ and satisfies $f(-2) = 7$. (As a bonus check, verify what happens at $x = 2$.)
14. Let $f(x) = \begin{cases} \frac{x^2 - 9}{x - 3}, & x \neq 3 \\ k, & x = 3 \end{cases}$. Find the value of k that makes f continuous at $x = 3$.
15. Let $f(x) = \begin{cases} \sqrt{x + 4}, & x \leq 0 \\ 2x + a, & x > 0 \end{cases}$. Find the value of a that makes f continuous at $x = 0$.
16. Let $f(x) = \begin{cases} x + 1, & x < 2 \\ 5, & x = 2 \\ x + 3, & x > 2 \end{cases}$. Determine whether f is continuous at $x = 2$. If not, classify the discontinuity (removable, jump, or infinite) and explain your reasoning.
17. Let $f(x) = \frac{1}{(x - 4)^2}$. Determine whether f is continuous at $x = 4$. If not, classify the discontinuity and explain your reasoning.

Applied

18. A simplified tax model gives tax owed (in dollars) as $T(x) = \begin{cases} 0.10x, & x \leq 50 \\ 0.12x - k, & x > 50 \end{cases}$, where x is income in thousands. Find the value of k that avoids a sudden jump in tax owed at $x = 50$ (i.e., makes T continuous there).
19. A shipping company's cost model is $C(w) = \begin{cases} 5 + 2w, & w \leq 3 \\ b + 3w, & w > 3 \end{cases}$ dollars, where w is weight in pounds. Find the value of b that makes the price transition smoothly (continuously) at $w = 3$.
20. A phone data plan costs $D(x) = \begin{cases} 20, & x \leq 1 \\ 20 + m(x - 1), & x > 1 \end{cases}$ dollars for x gigabytes. You're told that using 5 GB costs \$52. Find m . (As a bonus, notice that continuity at $x = 1$ holds automatically no matter what m is — explain why, by checking what the right-hand piece gives at $x = 1$.)
21. A temperature model is $T(t) = \begin{cases} 2t + 50, & t \leq 4 \\ -3t + c, & t > 4 \end{cases}$ degrees, where t is hours after midnight. Find the value of c that avoids a sudden jump in temperature at $t = 4$.